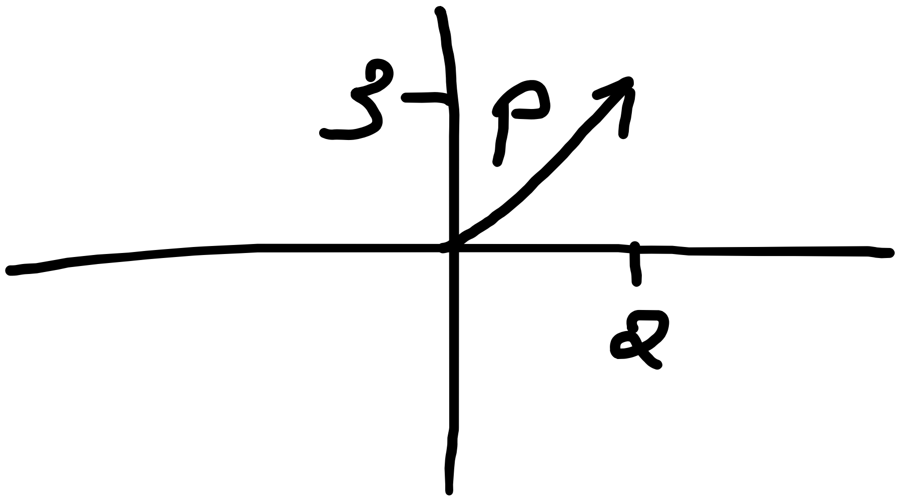
Introduction:



In our coordinate system, we would call this point $(2, 3)$ where the x-coordinate is 2 and the y-coordinate is 3.

In vector notation, this vector p has many representations:

$\binom{2}{3}$

$\overrightarrow{OP}$

p

$2i + 3j$

are the different notations used to represent the same vector.

We would write the vector like this:

$p = \overrightarrow{OP} = \binom{2}{3} = 2i + 3j$

Any of the two from $\overrightarrow{OP}$, p are acceptable for the left-hand side, and any of the two from $\binom{2}{3}$, $2i + 3j$ are acceptable for the right-hand side

Unit Vectors:

A unit vector is a vector with magnitude 1

Earlier we defined the vector p, which is a position vector, i.e., it defines the position of a vector in the plane

To find the respective unit vector $\hat{p}$, we must divide p by its magnitude:

$$|p| = \sqrt{2^2 + 3^2} = \sqrt{13}$$

$$\hat{p} = \frac{1}{\sqrt{13}}\binom{2}{3}$$

Operations involving vectors:

$$\binom{a}{b} + \binom{c}{d} = \binom{a+c}{b+d}$$

$$\binom{a}{b} - \binom{c}{d} = \binom{a-c}{b-d}$$

$$k\binom{a}{b} = \binom{ka}{kb}$$

Two vectors are parallel, if they are scalar multiples of each other

Resultant vectors: